

Giovanni Palla

Research Scientist, AI/ML, Chan Zuckerberg Initiative

Work

- 09/2024 **Research Scientist**, AI/ML, Chan Zuckerberg Initiative, Redwood City, California, present US.
- 04/2023 **Co-founder and Core Member**, scverse, Leading OSS ecosystem for single-cell present analysis in Python.
- 04/2024 **Bridging Post-Doc**, Prof Dr Fabian Theis's group, Machine Learning and
09/2024 Single-Cell Genomics, Institute of Computational Biology, Helmholtz Center
Munich, Munich, Germany.

Education

- 10/2019 **PhD student, awarded cum laude**, Prof Dr Fabian Theis's group, Machine Learning
04/2024 and Single-Cell Genomics, Institute of Computational Biology, Helmholtz Center
Munich, Munich, Germany.
- 09/2016 **MSc in Drug Innovation, 9.1/10 cum laude**, Graduate School of Life Sciences,
11/2018 Utrecht University, Utrecht, The Netherlands.
- 09/2013 **BSc in Biomolecular Sciences and Technology, 110/110 cum laude**, University of
07/2016 Trento, Trento, Italy.
- 06/2012 **High School Diploma, Scientific Field**, Liceo Scientifico G.B. Quadri, Vicenza,
Italy.

Publications

- 2025 A Large-Scale Benchmark of Cross-Modal Learning for Histology and Gene Expression in Spatial Transcriptomics, Gindra*, **Palla***, et al., ICCV 2025 CVAMD.
- 2025 Mapping cells through time and space with Moscot, Klein*, **Palla***, Lange*, Klein*, Piran* et al., 10.1038/s41586-024-08453-2.
Nature
- 2025 SpatialData: an open and universal data framework for spatial omics, Marconato*, **Palla***, Yamauchi*, Virshup* et al., 10.1038/s41592-024-02212-x.
Nature Methods
- 2023 The scverse project provides a computational ecosystem for single-cell omics data analysis, Virshup*, Bredikhin*, Heumous*, **Palla***, Sturm*, Gayoso* et al., 10.1038/s41587-023-01733-8.
Nature Biotechnology
- 2022 Modeling Single-Cell Dynamics Using Unbalanced Parameterized Monge Maps, Eyring*, Klein*, **Palla*** et al., Talk @ NeurIPS - LMRL Workshop 2022, 10.1101/2022.10.04.510766.
bioRxiv
- 2022 Uncertainty Quantification for Atlas-Level Cell Type Transfer, Engelmann*, Hetzel*, **Palla*** et al., Poster @ ICML 2022 - Workshop on Computational Biology, arXiv:2211.03793.
arXiv

- 2022 Squidpy: a scalable framework for spatial omics analysis, **Palla***, Spitzer* et al., 10.1038/s41592-021-01358-2.
Nature Methods
- 2022 Spatial components of molecular tissue biology, **Palla***, Fischer* et al., 10.1038/s41587-021-01182-1.
Nature Biotechnology
- 2020 Latent Factor Modeling of scRNA-Seq Data Uncovers Dysregulated Pathways in Autoimmune Disease Patients, **Palla*** et al., 10.1016/j.isci.2020.101451.
iScience

Conference Organization

- 2025 **Co-organizer**, 2nd scverse conference, Stanford, Palo Alto, California.
- 2025 **Co-organizer**, Learning Meaningful Representation of Life, ICML Workshop, Singapore.

Presentations, Lectures, and Conference Contributions

- 2023 **Lecturer**, Spatial Omics Data Analysis, MSc in Bioinformatics and Functional Genomics, IFOM.
Milan, Italy
- 2023 **Poster**, Mapping cells through time and space with Moscot, Single Cell Genomics 2023, Engelberg, Switzerland.
- 2023 **Poster**, SpatialData and Moscot for the integrative analysis of spatial omics data, Human Cell Atlas General Meeting 2023, Toronto, Canada.
- 2022 **Lecturer**, Spatial Omics Data Analysis, MSc in Bioinformatics, University of Verona, Verona, Italy.
- 2022 **Instructor**, Spatial Omics Data Analysis, SciLife Lab, 1-week course, Stockholm, Sweden - Virtual.
- 2022 **Instructor**, Advanced topics in single-cell analysis, Swiss Institute for Bioinformatics, 1-week course, Basel, Switzerland.
- 2022 **Invited talk**, ECCB 2022 - Workshop on Spatial Transcriptomics and cell-cell communication modeling, ECCB 2022, Sitges, Spain - Virtual.
- 2022 **Poster**, SpatialData: a FAIR framework for spatial multiomics data, Single Cell Genomics 2022, Utrecht, Netherlands.

Internships

- 06/2023 **Intern**, BioML team with Lorin Crawford, Microsoft Research, Cambridge, MA, USA.
08/2023
- 11/2018 **Intern**, Bioinformatics team with Enrico Ferrero, Autoimmunity, Transplantation
08/2019 & Inflammation, Novartis Institute for Biomedical Research, Basel, Switzerland.
- 11/2017 **Intern, Otto Bayer fellow**, Dr Judith Zaugg's group, Structural and Computational
08/2018 Biology Unit, European Molecular Biology Laboratory, Heidelberg, Germany.

Grants, Honors, and Awards

- 2023 **SpatialData: A FAIR Framework for Multimodal Spatial Omics**, CZI Single Cell Data Insight grant, Key personnel, PIs: Moore, Stegle.

- 2022 **Interactive Visualization of Spatial Omics Data**, napari Plugin Accelerator, Key personnel, PI: Theis.
- 2021 **Add On fellowship**, Joachim Herz Stiftung.
- 2017 **Otto Bayer Fellowship**, Bayer Foundation.
- 2017 **Erasmus Scholarship**, Utrecht University.
- 2016 **Young Innovators Honor Program**, Utrecht University.
- 2013 **Merit award for outstanding results in undergraduate studies**, University of Trento.

Teaching and Mentorship Experience

- 2020 **Team Leadership and Mentorship**, Spatialdata, Squidpy and Scverse teams.
- 2024
- 2020 **MSc and BSc Thesis Supervision**, Supervised 5 thesis students (4 MSc, 1 BSc).
- 2024

Journal and Conference Reviewing

- 2024 **Nature Communications**.
- 2023 **Optimal Transport and Machine Learning workshop**, NeurIPS.
- 2023 **Nature Methods**.
- 2021 **Bioinformatics**.

Extra-Curricular Activities

- 11/2019 **Student representative**, Munich School for Data Science, Munich, Germany.
- 12/2022 Interdisciplinary graduate program between Helmholtz Munich, TUM, LMU and DLR.
- 11/2016 **Board Member**, Club Alpbach Trentino, Trento, Italy.
- 09/2018 Part of the Forum Alpbach Network, European Forum Alpbach, Austria.
- 08/2016 **Scholarship holder**, European Forum Alpbach, Alpbach, Austria.
- 12/2015 **Board member**, OWL Open Wet Lab, MUSE (Science Museum), Trento, Italy.
- 12/2016 First Biohacking association in Italy. Responsible for the joint-editorial staff with local newspapers.
- 01/2015 **Science Communicator**, Ragionevolmente, Trento.
- 08/2016 Public events of science communication held in Trento and Bolzano.